

Market-Neutral Alpha in Crypto Perpetual Futures: Ranking Each Asset Against Itself

August 2026

Dhanya MD

Independent Researcher

Jayanagar, Hassan

Karnataka 543201, India

+91 93534 96617

dhanya13md@gmail.com

Dhanya MD is an independent quantitative researcher.
Corresponding author.

Market-Neutral Alpha in Crypto Perpetual Futures: Ranking Each Asset Against Itself

August 2026

ABSTRACT

Factor investing in equities works by comparing companies to one another. Shared accounting rules, industry membership and cash flow make those comparisons meaningful. Cryptocurrency perpetual futures share none of that, yet practitioners rank them against each other anyway. We test what happens when you stop. Applying ten factors drawn from published equity research to 112 crypto contracts over seven years, we find that scoring each contract against its own trailing history, rather than against its peers, raises average annual Sharpe from 1.01 to 1.45. The improvement holds for nine of the eleven factors tested and survives removing the funding component entirely. Building one portfolio per factor and blending returns, rather than blending scores into a single ranking, adds a further 0.67 Sharpe. The combined strategy returned 27% annually with a 7.6% maximum drawdown and gained 9% during 2022, when the market fell 82%.

KEY TAKEAWAYS

- Crypto contracts have no shared fundamentals, so ranking them against each other compares numbers that do not mean the same thing. Ranking each against its own history fixes this and adds 0.44 Sharpe.
- Funding is not a cost adjustment in perpetual futures. It is a return driver worth 40% to 266% of each factor's performance, and a price-only backtest understates results.
- Holding one book per factor and blending the returns beats blending signals into a single score by 0.67 Sharpe, with both approaches tuned over the same 1,286-configuration grid.

Keywords: market neutral, cryptocurrency, perpetual futures, factor investing, cross-sectional ranking, portfolio construction

JEL: G11, G12, G14, C58

Every cross-sectional strategy rests on an assumption that is rarely stated out loud: that the assets being ranked are comparable. When a portfolio manager sorts equities on book-to-price and buys the cheapest quintile, the sort is meaningful because a price-to-book ratio of 0.8 means roughly the same thing at one industrial company as at another. Shared accounting standards, industry classification and observable cash flow supply that common ground.

Cryptocurrency perpetual futures supply none of it. There is no accounting identity linking one token to another, no sector taxonomy with economic content, no cash flow to normalise by. A market-capitalisation figure depends on a token release schedule set by a founding team. Trading volume spans four orders of magnitude across contracts on the same exchange. Yet the strategies practitioners run in this market are almost uniformly cross-sectional: rank every contract on a signal, buy the top, sell the bottom.

The cross-sectional approach has deep roots in equities, from Fama and French (1993) through Asness, Moskowitz and Pedersen (2013), and it travels well across equity markets precisely because the comparability assumption holds there. Recent work extends factor pricing to digital assets (Cong et al. 2026), but does so by manufacturing a comparability anchor rather than questioning whether one exists.

This article asks what that assumption costs, and what happens if you abandon it. The alternative is straightforward. Instead of asking whether a contract's momentum is high relative to other contracts, ask whether it is high relative to its own history. Each asset becomes its own benchmark. The comparison no longer requires the assets to be alike, only that each has a past. The idea has a precedent in time-series momentum (Moskowitz, Ooi and Pedersen 2012), which judges each instrument against its own trailing return rather than against peers; we apply the same logic to the ranking step of a cross-sectional book.

We test this on 112 US-dollar-margined perpetual futures over 363 weeks, using ten factors taken from published equity research plus one built specifically for this market. The change in ranking frame is worth 0.44 in annual Sharpe ratio, holding in nine of eleven cases. A second construction choice, holding a separate portfolio for each factor rather than blending signals into one score, is worth a further 0.67. Together they turn a set of individually unremarkable signals into a strategy that returned 27% annually with a maximum drawdown under 8%.

The remainder of this article describes the market's peculiarities, the strategy, what each design decision contributes, and the evidence that the result is not an artefact of testing too many ideas against too little data.

WHY THIS MARKET IS DIFFERENT

Three features of perpetual futures matter for portfolio construction.

No anchor for comparison. The absence of shared fundamentals is not merely inconvenient. It means the same raw factor value carries different information at different contracts. A

volatility reading of 30% annualised is unremarkable for a small-cap token and extraordinary for the largest contract on the exchange. A cross-sectional sort treats those two readings as identical evidence. They are not.

Funding is a return, not a fee. Perpetual futures have no expiry. They track spot through a funding payment exchanged between long and short holders, typically every eight hours. Most practitioners treat this as a carrying cost to be netted out. That is a mistake here. Over our sample the funding component contributes between 40% and 266% of each factor’s total return, and a book constructed as ours receives funding in 98% of weeks. A backtest run on price alone understates this strategy rather than flattering it.

Aggressor-side information is published. Centralised crypto venues publish which side of each trade was the aggressor, information that equity researchers must infer with classification algorithms (Lee and Ready 1991). This makes it possible to measure something equities cannot observe directly: whether buy-initiated trades arrive in a different size distribution than sell-initiated ones. We introduce a factor built from the difference in skewness between the two. We are not aware of prior work that uses the skewness of the trade-size distribution, measured separately by initiating side and differenced over time, as a cross-sectional return predictor. Zhou (2008) partitions the size distribution by side in an order-driven equity market but measures tail exponents, finding them statistically indistinguishable across sides; a separate literature prices the skewness of *returns*. Neither measures the skewness of trade *sizes*, nor how the asymmetry between sides evolves.

The construction is simple enough to state in one line. Within each five-minute bar, classify the bar as buy-dominated or sell-dominated by aggressor volume, take the log of the average trade size in each, compute the skewness of each series over the week, and difference the two:

$$\text{TSKD}_t = \Delta \left[\text{skew}(\log \text{size}^B) - \text{skew}(\log \text{size}^S) \right]$$

The differencing matters: the level of the asymmetry is a persistent property of a contract, while the *change* in it is what carries information. The measure requires five-minute resolution and cannot be recovered from daily bars, which is why it is absent from the equity literature that relies on inferred trade direction.

The intuition is that informed and uninformed order flow have different size signatures. A participant with a view accumulates in a pattern that differs from retail flow reacting to a price move, and the difference shows up in the tail of the size distribution rather than in its average. Averages are what order-imbalance measures capture, and they discard exactly the information that distinguishes the two. Measuring the skewness gap between buy-dominated and sell-dominated intervals retains it.

THE FACTORS

Eleven factors are used. Ten are transported from published equity research and the formulaic-alpha literature (Kakushadze 2016) rather than invented here, which is deliberate: the article’s

claim concerns how factors are ranked and combined, not whether we can find new ones. Using established signals means any improvement is attributable to construction rather than to signal discovery.

They fall into four families. *Volatility and jump measures* capture the character of recent price movement, distinguishing realised upside from downside variation (Barndorff-Nielsen, Kinnebrock and Shephard 2010) and separating continuous moves from jumps (Patton and Sheppard 2015). *Liquidity and impact measures* relate price movement to the volume required to produce it, including a spread proxy and a quadratic-impact term. *Order-flow measures* use the aggressor-side data described above, including signed imbalance (Chordia, Roll and Subrahmanyam 2002) and the trade-size asymmetry factor; the premise that order size carries information traces to Kyle (1985). *Positioning measures* use exchange-published statistics on open interest and the long-short ratios of top traders relative to the broader account base.

Individually, none of these is remarkable. Traded alone in the conventional cross-sectional frame, the eleven average an annualised Sharpe of 1.01, which after costs is not a business. The argument of this article is that most of the value in this market sits in decisions taken after the signals are chosen.

Exhibit 1 lists them with the data each requires.

Exhibit 1: The Eleven Factors and Their Data Requirements

Family	Factor	Data required
Volatility and jumps	Realised semivariance ratio	daily bars
	Signed jump variation	daily bars
	Adjusted volatility	daily bars
Liquidity and impact	Spread proxy	daily bars
	Quadratic impact	daily bars
	Volume-scaled impact, mean	daily bars
	Volume-scaled impact, variance	daily bars
Order flow	Order-flow imbalance	daily bars
	Trade-size skewness gap	5-minute
Positioning	Open-interest change	exchange metrics
	Top-trader vs. all-account ratio	exchange metrics

Ten of the eleven are transported from published equity research. Only the trade-size skewness gap requires intraday resolution; a practitioner working from daily bars can implement the other ten. Exchange metrics are published free and require no credentials.

A practitioner without five-minute history can implement ten of the eleven and lose relatively little. The trade-size factor contributes positively but is not load-bearing.

THE STRATEGY

The construction has four steps.

Score each contract against its own past. For each of eleven factors, a contract's raw factor value in a given week is converted to a percentile against that contract's own trailing 52-week distribution of the same factor. The score is centred so that zero is the contract's own median. Nothing in the calculation refers to any other contract. Critically, the trailing window uses only weeks strictly before the scoring week, so a score can be computed in real time with no knowledge of the future.

To see the difference this makes, consider two contracts in a single week of our sample. One shows a raw factor value of -27.86 , the lowest in the cross-section. The other shows -11.74 , far higher. Cross-sectional ranking would buy the second and sell the first. Measured against their own histories, the first is only mildly unusual and scores near the middle, while the second is extreme for that contract and scores near the top. The self-referential frame reverses the conclusion, and it is the frame that proves correct.

Build one portfolio per factor. Each factor independently sorts the tradeable universe and holds the top 20% long and the bottom 20% short, equally weighted, dollar neutral. With 112 contracts that is 22 positions a side at roughly 4.5% each. The quintile fraction is a parameter, not a constant: it is selected on historical data alongside everything else, and on a smaller universe the position count shrinks with it.

Combine returns, not scores. The eleven single-factor portfolios are combined by weighting each in inverse proportion to its own realised volatility, so a steadier factor contributes more. Combining weakly correlated books rather than collapsing them into one is what Grinold's (1989) fundamental law would predict: information ratio scales with the square root of the number of independent bets, and blending scores destroys that breadth. This is the step most implementations skip. The common approach averages the eleven scores into one composite and sorts on that. Averaging scores destroys the information that distinguishes factors from one another; a contract that one factor loves and another hates ends up indistinguishable from a contract both are indifferent to.

The difference is easiest to see in the final weights. In a representative week of our sample, one contract is shorted by six of the eleven factors and held by none, and ends with a portfolio weight of -2.3% . Another is shorted by four and held by one, ending at -1.5% . A third, shorted by two and held by one, ends at -0.7% . The position sizes carry information about how much the factors agree. A blended score would have compressed all three into a single ranking and sized them by their position in that ranking, discarding the distinction between broad agreement and marginal disagreement.

The volatility weighting matters for a second reason. The eleven factors have materially different return volatilities, and equal-weighting their books would let the noisiest one dominate the combined portfolio's risk regardless of its contribution. Weighting by inverse volatility equalises

risk contribution instead of capital, which is standard practice in multi-strategy allocation and is unusually valuable here because the dispersion in factor volatility is wide.

Tilt toward carry and cap turnover. Positions are tilted toward receiving funding rather than paying it, and weekly turnover is capped. Average turnover runs at 37% of gross exposure per week.

Universe and rebalancing. The tradeable universe is screened on trailing dollar volume, which removes contracts too thin to trade at size without moving the price. The screen is applied on trailing data only, so a contract that becomes liquid enters the universe when it does so and not retrospectively. This detail matters more than it sounds: applying a liquidity screen using full-sample data is one of the most common ways a crypto backtest quietly acquires hindsight, because the contracts that survive to be liquid at the end of the sample are exactly the ones that did well.

Rebalancing is weekly. That frequency is a compromise: the factors decay over several weeks, so daily rebalancing adds cost without adding information, while monthly rebalancing gives up too much of the signal. Weekly also happens to be long enough that execution can be worked patiently rather than taken at market, which is what makes the cost assumptions in this article realistic.

WHAT EACH CHOICE CONTRIBUTES

Exhibit 2 isolates the ranking frame. Every other element is held fixed and only the comparison basis changes.

Exhibit 2: The Ranking Frame

	Cross-sectional	Self-referential	Factors improved
Mean Sharpe, with funding	1.011	1.451	9 of 11
Mean Sharpe, funding removed	0.383	0.648	8 of 11

Mean annualised Sharpe ratio across eleven single-factor portfolios, 112 contracts, 363 weeks. The second row removes the funding component entirely, showing the effect is not a carry artefact.

The improvement is not explained by funding: with the funding component removed entirely, the ranking frame still adds 0.27 and still holds in eight of eleven cases. Nor is it a statistical curiosity. Standardising each contract against its own history before ranking cross-sectionally, an intermediate step that removes scale differences but keeps the peer comparison, recovers 41% of the gap. That identifies scale heterogeneity as one mechanism and leaves the remainder to the frame itself.

It is worth being precise about what that 41% means, because it points at what is actually going wrong in the conventional approach. When a cross-sectional sort ranks contracts on a raw factor value, it is implicitly assuming the factor has the same distribution everywhere. It does not.

A liquidity measure computed on the largest contract and on a token trading a thousandth of its volume produce numbers on entirely different scales, so the sort ends up ordering contracts by size rather than by the characteristic it intended to capture. Standardising each contract first removes that contamination, and recovering 41% of the gap tells you scale was roughly half the problem.

The other half is harder to remove, and it is the part that motivates abandoning the peer comparison altogether. Even after standardisation, a cross-sectional sort still asks which contract is most extreme relative to the others this week. In a market where the constituents genuinely differ, that question has no stable answer, because the composition of the cross-section changes as contracts list, delist and move in and out of the liquidity screen. Asking instead whether each contract is extreme relative to its own past sidesteps the composition problem entirely.

Exhibit 3 isolates the construction choice, comparing one-book-per-factor against the conventional blended-score pipeline. Both were tuned over an identical grid of 1,286 configurations, so neither is handicapped.

Exhibit 3: Portfolio Construction

	Best configuration	Average configuration
One book per factor, self-referential	2.571	1.537
Blended scores, cross-sectional	1.971	0.995
Mean difference, paired ($t = 5.82$)	0.665	—
Matched cells favouring one book	15 of 17	—

Annualised Sharpe ratio. The paired comparison holds every parameter fixed and varies only construction, so the difference is attributable to construction alone.

The ordering holds on both the best configuration and the average one, which matters: a result that appears only at the optimum is usually a result about the optimiser.

Carry deserves separate mention. Traded on its own the funding component returns an annualised Sharpe of 1.25 ($t = 2.91$), respectable but short of the $t > 3.0$ bar and so not by itself compelling. Its value is as a component. Removing it costs the least-affected factor about 40% of its performance and the most-affected more than two-thirds. Practitioners who net funding out of their backtests as a cost are discarding a return source.

PERFORMANCE

Exhibit 4 summarises the combined strategy.

Exhibit 4: Strategy Performance, 282 Weekly Rebalances

Cumulative return	+327.6%
Annualised return	+27.8%
Annualised volatility	12.8%
Sharpe ratio	2.16
<i>t</i> -statistic	5.03
Maximum drawdown	−7.6%
Recovery from maximum drawdown	16 weeks
Worst week	−6.2%
Positive weeks	65.5%
Return to maximum drawdown	3.7
Skewness	+1.14
Weekly turnover	37%

The *t*-statistic of 5.03 is worth pausing on. Harvey, Liu and Zhu (2016) argue that the flood of published anomalies means a newly claimed factor should clear $t > 3.0$ rather than the conventional 2.0. This result clears that raised bar by two-thirds, on 282 independent weekly observations.

Two features deserve attention beyond the headline. The maximum drawdown of 7.6% recovered within sixteen weeks, giving a return-to-drawdown ratio of 3.7. And returns are positively skewed at +1.14. Most systematic strategies that produce high Sharpe ratios do so by selling tails and exhibit negative skew; this one does the opposite, which matters for anyone sizing the position.

The strategy's usefulness rests on a claim of market neutrality. A regression of weekly returns on market, size and momentum factors built from the same universe is reported in Exhibit 5. The market loading is -0.001 with a *t*-statistic of -0.12 : the book is market-neutral *empirically*, not merely by construction. Size and momentum load negatively but explain little, and the regression R^2 is 0.074. The intercept is an annualised alpha of 27.05% with a *t*-statistic of 4.42 after Newey–West (1987) correction, accounting for 97% of the strategy's mean return. Three factors that explain much of what happens in this market explain almost none of this.

Exhibit 5: Is It Alpha? Regression on Market, Size and Momentum

	Coefficient	<i>t</i> (OLS)	<i>t</i> (NW)
Alpha	0.00520	5.03	4.42
Market	−0.00128	−0.13	−0.12
Size	−0.09260	−3.83	−2.69
Momentum	−0.06819	−3.39	−2.23
Annualised alpha			+27.05%
R^2			0.074
Alpha share of mean return			97%
Observations			281 weeks

Weekly strategy returns regressed on an equal-weighted market factor, a size book and a twelve-week momentum book, all built point-in-time from the same universe. Newey–West standard errors correct for autocorrelation; first-order autocorrelation is +0.063, so the correction is modest.

An average beta of zero over seven years does not by itself establish that the book is neutral when it matters. Exhibit 6 tests it in the period most likely to break it.

Exhibit 6: The 2022 Decline, 61 Weeks

	Market	Strategy
Cumulative return (%)	−82.5	9.4
Maximum drawdown (%)	−84.0	−7.6
Correlation over window	0.045	

Market is the equal-weighted return of the tradeable universe, November 2021 to December 2022.

IS IT REAL?

Any strategy reported after searching a parameter space needs to answer why the result is not the best of many tries. We ran four tests, and report the ones that failed alongside the ones that passed.

Selection out of sample. Configuration is chosen on a trailing two-year window and evaluated on the following six months, rolling forward. Every parameter is reselected in each window, including the quintile fraction; holding any parameter at its full-sample best would let the test inherit knowledge of the evaluation period. Because a weekly strategy also carries hidden exposure to which weekday it rebalances (Hoffstein, Sibears and Faber 2019), we rebuilt the entire dataset on each of seven rebalance weekdays and repeated the exercise. We are not aware of these two procedures being applied together before. Separately each leaves a gap: a walk-forward result still inherits an arbitrary rebalance weekday, and a timing-luck study still inherits a fixed configuration. Running them jointly closes both at once.

Average out-of-sample Sharpe across the seven schedules is 1.87, and all seven are prof-

itable. Five of the seven clear a t -statistic of 3.0, the threshold Harvey, Liu and Zhu (2016) propose for claimed anomalies. Two do not, and we report them as failures rather than reporting the best weekday and stopping. The dispersion in Exhibit 7 is instructive on its own: a study quoting a single schedule could have reported anything between 1.07 and 2.58 based on a choice with no economic content.

Exhibit 7: Out-of-Sample Result by Rebalance Weekday

Rebalance day	Sun	Mon	Wed	Sat	Tue	Fri	Thu
Out-of-sample Sharpe	2.58	2.54	1.89	1.84	1.70	1.46	1.07
t -statistic	4.82	4.75	3.53	3.44	3.17	2.73	2.01
Clears $t > 3.0$	yes	yes	yes	yes	yes	no	no

182 out-of-sample weeks per schedule. Average across the seven is 1.87. The entire dataset is rebuilt from hourly bars on each weekday, so the seven are genuinely independent schedules rather than shifted views of one. Monday, the schedule we trade, is second of seven rather than first.

Generalisation across assets. Time-series testing does not show that parameters transfer to contracts they were not fitted on. We selected configurations on 78 contracts and evaluated on the 34 held out. All eight splits were profitable, retaining 83% of in-sample performance once the comparison controls for the fact that a smaller universe holds fewer positions.

Correcting for the search itself. Every configuration evaluated during development was logged to an append-only record, so the number of trials is counted rather than recalled. Applying the deflated Sharpe ratio (Bailey and López de Prado 2014) against the full logged count of 1,286 trials, the probability that the result reflects skill rather than selection is 0.95, short of the 0.99 threshold we set in advance. Against the narrower count of 211 configurations in the final search it is 0.99. We report both, including the one that fails.

Costs. Transaction costs are modelled per contract on trailing dollar volume, at 5 to 25 basis points round trip. The strategy remains profitable at eight times the modelled level. That is the failure mode most likely to be understated in a backtest, so it deserves a wide margin.

Exhibit 8 collects the results.

Exhibit 8: Robustness Summary

Test	Result	Threshold
Walk-forward, seven rebalance weekdays	Sharpe 1.87	> 0
individual weekdays clearing $t > 3.0$	5 of 7	—
Held-out contracts, eight splits	83% retained	$> 50\%$
Deflated Sharpe, 211 trials	0.99	0.99
Deflated Sharpe, 1,286 trials	0.95	0.99
Transaction costs	profitable at $8\times$	$1\times$
Market loading (Exhibit 5)	-0.001 ($t = -0.12$)	≈ 0

Thresholds are stated in advance rather than chosen after the fact. Two of the seven rebalance weekdays fall short of $t > 3.0$, and the deflated Sharpe computed against the full logged trial count falls short of 0.99; both are shown here rather than omitted.

DOES THE LIVE BOOK RUN THE BACKTEST?

A backtest describes a system that does not exist. The system that eventually trades is a separate piece of software, usually written later, often by someone else, and the assumption that the two compute the same positions is almost never tested. We are not aware of a published strategy paper that verifies it.

We do, with an automated gate that runs before any deployment. Research and the live path call the same scoring module, and the gate asserts that they produce identical weights, contract by contract, at every rebalance in the sample. It also re-tests every factor for look-ahead by recomputing a historical score with all subsequent data deleted and requiring an exact match, checks that the book is dollar neutral, and fails if a configuration produces no positions at all. It exits non-zero on any violation, so a broken build cannot ship.

The gate matters because it has caught real failures. An earlier version of this work shipped a live book that had drifted to a rank correlation of 0.15 against research, with position overlap barely above chance. A later look-ahead bug in a shared ranking helper survived undetected precisely because research and production agreed perfectly: both called the same incorrect function. That second failure is the instructive one. Agreement between two systems is not evidence of correctness if both inherit the same mistake, which is why the gate checks point-in-time integrity independently rather than only comparing the two paths to each other.

For a practitioner the implication is direct. The claim that a live system implements a published backtest is an empirical proposition, not a safe assumption, and it is cheap to test.

WHAT THIS DOES NOT SHOW

Three limitations are worth stating plainly, because a reader deciding whether to act on this should know where the evidence thins out.

The sample covers seven years of a single asset class in what has been, on net, a period of extraordinary growth and volatility. Seven years is a reasonable sample in weekly observations but a short one in market regimes. The 2022 decline is the only severe drawdown in it, and while the strategy came through that episode well, one episode is one observation.

Capacity is genuinely limited. At roughly \$67 million the strategy is a viable book for an individual or a small fund and irrelevant to a large one. Readers at institutions should treat the ranking-frame result as the transferable finding and the specific strategy as a demonstration of it.

Finally, the result is about construction, not forecasting. We have not shown that these eleven factors predict returns better than others, or that the effect persists as more capital pursues it. What we have shown is that given a set of signals, two decisions taken after signal selection are worth more than the signals themselves. That is a claim about method, and it should be tested in other markets that lack a natural peer group before it is treated as general.

COSTS, CAPACITY AND IMPLEMENTATION

The strategy trades 112 contracts weekly with 37% turnover, which is modest for a systematic book. All inputs are free and public: prices, volumes and funding rates come from an exchange bulk archive, positioning statistics from a public endpoint requiring no credentials. No paid data is used anywhere in the construction.

Capacity is the binding practical constraint. Assuming participation at 2% of each contract's trailing dollar volume, the strategy supports roughly \$67 million. That is a meaningful number for an individual manager or a small fund and an irrelevant one for a large institution. It also implies the effect is unlikely to be arbitrated away quickly, since the capital required to do so exceeds what the market will absorb.

One factor requires five-minute data to construct. Practitioners working from daily bars can implement ten of the eleven; the remaining factor contributes positively but is not load-bearing.

CONCLUSION

Cross-sectional ranking is so standard that its precondition goes unexamined. That precondition, that the assets being compared are genuinely comparable, is supplied in equities by accounting and industry structure. Crypto perpetual futures do not supply it, and ranking them against each other imposes a comparison the market does not support.

Ranking each contract against its own history removes the requirement. It is not a refinement of cross-sectional investing but a different premise, and in this market it is worth 0.44 in Sharpe across eleven independent factors. Holding factors in separate portfolios rather than blending their signals is worth a further 0.67. Neither choice requires new data, a proprietary signal or a forecasting model. Both are decisions about construction, made before any factor is chosen.

For a practitioner the practical implication is narrower and more useful than the strategy itself. If you already run a cross-sectional book in this market, the ranking frame is a change you can test in an afternoon against your existing signals: replace the cross-sectional percentile with a trailing self-percentile and rerun. It requires no new data and no new signal research, and in our sample it was worth more than any individual factor we tested. The construction change, holding separate books rather than blending scores, is a larger refactor but is similarly independent of which signals you use.

The broader implication is that in markets without a natural peer group, the comparison basis is a design decision rather than a convention to inherit. Equity practice supplies the cross-sectional default, and it supplies it for a good reason, because equities have the shared structure that makes peer comparison meaningful. Carrying that default into a market that lacks the structure imports the machinery without the justification. The same question arises anywhere the constituents are genuinely heterogeneous: commodities across storability regimes, credit across capital structures, private assets with no common valuation basis. The comparison basis deserves the same scrutiny practitioners already give to signal selection, and in markets like this one it matters more.

DECLARATION OF SUBMISSION

The author has read the publication agreement and is aware of the conditions of submission. The author declares that no institution requires that this paper be posted on any public-facing website, including SSRN, ResearchGate or SciHub.

DECLARATION OF INTEREST

The author declares no competing financial or personal interests. This research received no funding. The author has no affiliation with, consulting relationship with, or financial interest in any cryptocurrency exchange, trading venue or data provider referenced in this study.

REFERENCES

- Asness, Clifford S., Tobias J. Moskowitz, and Lasse Heje Pedersen. 2013. “Value and Momentum Everywhere.” *The Journal of Finance* 68 (3): 929–985.
- Bailey, David H., and Marcos López de Prado. 2014. “The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality.” *The Journal of Portfolio Management* 40 (5): 94–107.
- Barndorff-Nielsen, Ole E., Silja Kinnebrock, and Neil Shephard. 2010. “Measuring Downside Risk: Realised Semivariance.” In *Volatility and Time Series Econometrics*, 117–136. Oxford: Oxford University Press.

- Chordia, Tarun, Richard Roll, and Avanidhar Subrahmanyam. 2002. "Order Imbalance, Liquidity, and Market Returns." *Journal of Financial Economics* 65 (1): 111–130.
- Cong, Lin William, G. Andrew Karolyi, Ke Tang, and Weiyi Zhao. 2026. "Crypto Value, Factor Pricing, and Market Segmentation." *Management Science*, forthcoming.
- Fama, Eugene F., and Kenneth R. French. 1993. "Common Risk Factors in the Returns on Stocks and Bonds." *Journal of Financial Economics* 33 (1): 3–56.
- Grinold, Richard C. 1989. "The Fundamental Law of Active Management." *The Journal of Portfolio Management* 15 (3): 30–37.
- Harvey, Campbell R., Yan Liu, and Heqing Zhu. 2016. "... and the Cross-Section of Expected Returns." *The Review of Financial Studies* 29 (1): 5–68.
- Hoffstein, Corey, Justin Sibears, and Nathan Faber. 2019. "Rebalance Timing Luck: The Difference Between Hired and Fired." *The Journal of Index Investing* 10 (1): 27–36.
- Kakushadze, Zura. 2016. "101 Formulaic Alphas." *Wilmott* 2016 (84): 72–81.
- Kyle, Albert S. 1985. "Continuous Auctions and Insider Trading." *Econometrica* 53 (6): 1315–1335.
- Lee, Charles M. C., and Mark J. Ready. 1991. "Inferring Trade Direction from Intraday Data." *The Journal of Finance* 46 (2): 733–746.
- Moskowitz, Tobias J., Yao Hua Ooi, and Lasse Heje Pedersen. 2012. "Time Series Momentum." *Journal of Financial Economics* 104 (2): 228–250.
- Newey, Whitney K., and Kenneth D. West. 1987. "A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix." *Econometrica* 55 (3): 703–708.
- Zhou, Wei-Xing. 2008. "Universal Price Impact Functions of Individual Trades in an Order-Driven Market." *Quantitative Finance*. Working paper.
- Patton, Andrew J., and Kevin Sheppard. 2015. "Good Volatility, Bad Volatility: Signed Jumps and the Persistence of Volatility." *The Review of Economics and Statistics* 97 (3): 683–697.